

Quality-Constrained Compression of Multi-Vector Retrievers:

Compression Axes, Certification Difficulty, and the Boundaries of What Generalizes

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Abstract—Multi-vector late-interaction retrievers store hundreds of vectors per page, which makes their indexes large. We study post-hoc, training-free compression of fixed page embeddings along three axes: vector count (Ward merging with per-page budgets), dimension (PCA) and quantization (int8, centred int4, binary). We evaluate four visual encoders (ColPali v1.3, ColQwen2 v1.0, ColQwen2.5 v0.2 and the 2,560-dimensional ColEmbed 4B) on ViDoRe V1 DocVQA and InfoVQA, plus a text ColBERT on SciFact, with pre-registered hypotheses. Within this tested set, fewer stored vectors is the only axis that reduces exhaustive search cost, which we measured to be linear in stored vectors over the tested range. Merging to a quarter of the vectors keeps nDCG@5 within 1.1 points of float32 on every 128-dimensional dataset measured, and merging beats random pruning. Per-vector int8 and centred int4 are near-lossless on every encoder measured, whereas dimension reduction is model-dependent: harmful at 128 dimensions, nearly free to one eighth of the width at 2,560. The axes interact: compression factors do not multiply, and PCA followed by coarse codecs is consistently antagonistic. DocVQA admits far less certified compression than InfoVQA mainly because its quality is harder to certify, not because it is less compressible. To select a configuration with a statistical statement, we apply known fixed-sequence testing with exact betting p-values. This bounds the probability of deploying a configuration below target by δ under i.i.d. queries and a fixed corpus, and the bound held in 27 of 27 frozen held-back cells. No distribution-free method can certify a 0.99 retention target with 225 or fewer calibration queries on these pools. Native compressed-code scoring was slower than decode-then-BLAS on our software stack, and retrieval quality beyond 500 pages is unmeasured. We report these findings as established within the tested scope; their novelty relative to the full literature is not established.

Index Terms—multi-vector retrieval, late interaction, embedding compression, quantization, token pooling, visual document retrieval, post-selection inference

I. Introduction

Late-interaction retrievers such as ColBERT [1], [2] and its visual descendants ColPali and ColQwen [3] represent each document by many vectors and score a query with MaxSim: for every query token, the best-matching document vector, summed over tokens. On page images this means hundreds of 128-dimensional vectors per page. A

float32 ColPali page is 528 KB, and a 2,560-dimensional ColEmbed 4B page is 7.8 MB. Storage, memory and exhaustive search cost all grow with that footprint.

Post-hoc, training-free compression offers several independent-looking axes. We can store fewer vectors per page (merging or pruning), narrower vectors (projection), or fewer bits per value (quantization). Each is often reported on its own, and combined savings are often quoted as the product of the individual factors. This paper does not propose a new compression algorithm. It asks five empirical questions about quality-constrained compression, and reports where each answer holds and where it does not:

- 1) Which axes actually reduce storage, memory and search cost?
- 2) How do the axes interact when combined?
- 3) Why do some datasets admit much less compression than others?
- 4) Can the selection of a compression configuration be statistically certified?
- 5) Which findings generalize across models and corpora?

Contributions. We separate research findings from engineering explicitly.

- **Research:** an empirical map of three compression axes and their interactions for multi-vector visual retrievers. It uses pre-registered hypotheses and paired confidence intervals, and states where each finding generalizes: four visual encoders and one text encoder on three corpora (Sections V and IX).
- **Research:** an explanation of dataset-dependent compression through certification difficulty, driven by the relevance-margin composition of the query pool, rather than through intrinsic compressibility (Section VI).
- **Methodology:** a validated application of known post-selection testing to compression selection: fixed-sequence testing with exact betting p-values [20], [23],

together with its finite-sample limits (Section VII). This is not a new statistical method.

- Engineering: an open, model-agnostic compression and selection framework (OptiVision v0.3.0). It provides composable stages, exact byte accounting, two-tier indexing and an exact batched rescoring path (Section III).

A final synthesis classified every finding as research, methodology, engineering or an implementation-specific observation. It concluded that the empirical findings are established within the tested scope, but that their priority or novelty relative to the complete literature is not. We keep that boundary throughout.

II. Background and Problem Formulation

Late interaction. A query q with token vectors $\{x_t\}$ and a page p with vectors $\{d_v\}$ score $s(q, p) = \sum_t \max_v x_t^\top d_v$. Exhaustive search over N pages costs work proportional to the number of stored vectors times the dimension.

Compression axes. All axes act on stored page vectors after encoding, without retraining:

- (A) vector count: Ward clustering within each page to a fixed fraction of its vectors, with each cluster represented by its mean rescaled to the members' mean norm. This is Ward token pooling [7], [8]. Protected non-patch tokens are kept unchanged.
- (B) dimension: uncentred PCA fitted on document vectors; queries are projected with the same map.
- (C) quantization: float16, per-vector int8 (one float16 scale per vector), plain and mean-centred int4, a 2-bit Lloyd codec, and 1-bit sign codes.

Quality constraint. For a configuration with per-query compressed and baseline scores c_i, b_i (nDCG@5, or nDCG@10 for SciFact), retention is $R = \sum_i c_i / \sum_i b_i$. A target T (0.99, 0.97 or 0.95) defines feasibility. Because

$$R \geq T \iff \mathbb{E}[c_i - T b_i] \geq 0, \quad c_i - T b_i \in [-T, 1], \quad (1)$$

certifying a target is a one-sided test on the mean of a bounded variable. This identity is what makes finite-sample, distribution-free testing applicable.

III. The OptiVision Framework (Engineering)

OptiVision composes token, dimension and codec stages into pipelines, fits each stage on document vectors only, and records actual stored bytes: codes, per-vector scales, offsets and shared state such as the PCA basis. Budgets are relative to each page (for example, merge to 1/4 of its vectors) because absolute thresholds do not transfer across encoders (Section IX).

Search. ExactIndex scores exact MaxSim over any store, decoding codes blockwise to float32 under a fixed memory budget. TieredIndex shortlists 50 pages with a small hot tier (binary codes in RAM) and rescores them with a cold tier (int8 codes). We rewrote the rescoring as a batched per-candidate-page computation. It reproduces the previous per-query implementation's rankings and

scores bit for bit on all six tested datasets (60 of 60 cells). It is an engineering optimization: median rescoring speedup $6.7\times$ and median end-to-end speedup $3.25\times$ on one laptop CPU, with the gain shrinking as candidate overlap falls (Section VIII).

Selection. `calibrate()/optimize()` search a 40-configuration default space and choose the smallest configuration whose one-sided 0.999 bootstrap lower bound on retention clears T . This is the shipped rule. It is empirically safe but has no post-selection guarantee (Section VII). The framework is released as an open package (v0.3.0).

IV. Experimental Methodology

Encoders and data.

- Visual encoders: ColPali v1.3 (vidore/colpali-v1.3-merged), ColQwen2 v1.0 and ColQwen2.5 v0.2 [3] (all 128-dimensional), and ColEmbed 4B (nvidia/nemotron-colembd-v1-4b-v2, 2,560-dimensional) [4].
- Visual data: ViDoRe V1 DocVQA and InfoVQA test splits [3], each 500 pages with 451 and 494 labelled queries.
- Text encoder: answerai-colbert-small-v1 (96-dimensional) on BEIR SciFact [5] (5,183 documents, 300 queries).

For ColEmbed only aggregate results with paired bootstrap intervals exist (no per-query data), and its timings are GPU timings.

Protocol. Each study froze its hypotheses, configurations and decision rules in a committed plan before any result was inspected. Other safeguards:

- Reported quantities: retention with paired bootstrap 95% intervals (1,000 resamples).
- Reproduction checks: any configuration shared with an earlier store had to reproduce its per-query scores exactly.
- Held-back confirmation: ColQwen2.5 and SciFact were withheld from the certification study until its method was frozen.
- Timing: laptop CPU (Intel i7-1165G7, numpy/OpenBLAS), batch-amortized over all queries, one untimed warm-up run and five timed rounds. A run interrupted by system standby was discarded and repeated with sleep disabled.

Evaluation alignment. OptiVision's relevance labels differ from the official ViDoRe evaluation [6] only for questions repeated on several pages. Aligning with the official legacy and MTEB labelling changed baselines by at most 0.5 points and retention by at most 0.65 points (Section X).

V. Compression Axes and Their Interactions

Table I and Fig. 1 summarize the single-axis behaviour. All retentions are measured against the same encoder's float32 index on DocVQA and InfoVQA.

TABLE I

Single-axis and combined compression (measured). Retention is nDCG@5 relative to the same encoder’s float32 index. “128-d” gives the range over ColPali, ColQwen2 and ColQwen2.5 on DocVQA and InfoVQA (six datasets); “2,560-d” is ColEmbed 4B (DocVQA / InfoVQA). Scan time is the median CPU exhaustive scan as a fraction of float32 (DocVQA, 128-d). Factors are actual stored bytes relative to float32 codes.

configuration	128-d retention	2,560-d retention	scan time / float32 (CPU, 128-d)
merge to 1/4 of vectors ($\approx 3.9\times$)	0.989–1.006	0.988 / 0.995	0.27–0.28
merge to 1/10	0.954–0.992	0.984 / 0.996	0.13
PCA to $d/2$	0.972–0.991	1.002 / 0.999	0.79–0.80
PCA to $d/4$	0.803–0.907	0.993 / 0.996	0.74
int8 per vector ($3.9\times$)	0.998–1.001	0.997 / 1.002	1.01–1.02
centred int4 ($7.8\times$)	0.990–1.001	0.997 / 1.002	1.03–1.04
plain int4 ($7.8\times$)	0.975–1.004	0.009 / 0.004	1.04–1.05
binary ($32\times$)	0.963–0.995	0.989 / 0.998	1.01–1.02
merge 1/4 + binary ($\approx 122\text{--}125\times$)	0.954–0.982	0.974 (DocVQA)	0.28
two-tier: merged binary in RAM + int8 rescoring of 50	0.999–1.005	0.997 / 1.003	RAM 122–125 $\times$ smaller

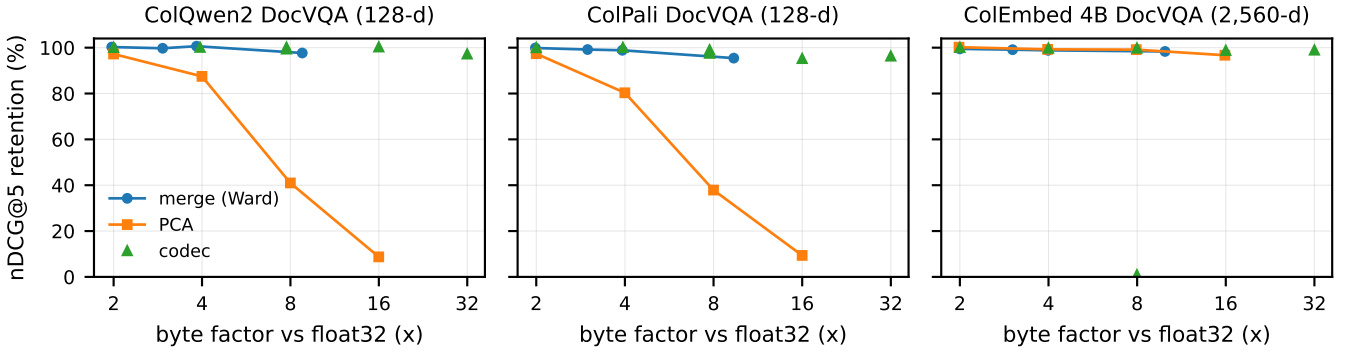


Fig. 1. Retention against byte factor for each axis on DocVQA (measured). Vector merging and per-vector codecs stay near float32 on all encoders. PCA collapses at 128 dimensions but not at 2,560. Plain int4 collapses on ColEmbed 4B (the codec point near zero at $8\times$, right panel).

Vector count drives search cost. Exhaustive scan time fell to 0.52–0.53, 0.27–0.28 and 0.16 of float32 at 1/2, 1/4 and 1/8 of the vectors on all three 128-d encoders. ColEmbed’s within-device (GPU) ratios behave the same way. Halving or quartering the width left 79–80% and 74% of the scan time. In the current implementation, codecs made scans 1–7% slower, because codes are decoded to float32 before the product. That is a property of the implementation, not of quantization (Section VIII).

Merging is the quality-preserving vector reduction. Merging to 1/4 of the vectors cost at most 1.1 points on every 128-d dataset. A controlled attribution at equal vector counts on six datasets isolated two effects. Ward clustering retains +0.7 to +3.7 points more than K-means with plain means, and rescaling merged vectors to their members’ mean norm adds a further +0.1 to +4.6 points. All merging variants beat random pruning in 51 of 54 comparisons, by up to 18 points at 1/20 of the vectors.

Codecs. Per-vector int8 and centred int4 stayed within about one point on every encoder measured, including the text model (centred int4 98.6%). Plain int4, 2-bit and binary codes are model-dependent: plain int4 collapses on ColEmbed and loses 10 points on the text model.

Compression factors do not multiply, and the axes

interact. The actual byte factor of a combination was 0.69–1.00 of the product of its single-axis factors at 128-d, because per-vector scales do not shrink with width. It was 0.53–1.00 on ColEmbed, where the shared PCA basis also counts. Retention interactions were measured as $I = R_{X+Y} - R_X R_Y$ with paired intervals:

- 14–34% of two- and three-axis combinations showed a material interaction ($|I| \geq 0.01$, interval excluding 0);
- every material interaction at 128-d was negative;
- they concentrated in PCA combined with binary or centred int4, reaching -18 points, with the same direction in ColEmbed’s point estimates;
- merging combined with any codec was close to multiplicative.

The practical consequence is that axes must be evaluated jointly, not tuned as independent knobs.

Width changes the best storage pipeline. On the in-sample storage frontier (members with nDCG@5 retention ≥ 0.90 , all queries), codecs alone cover the first $\approx 8\times$ at 128-d. Merging plus a codec then covers 8–125 $\times$, and PCA enters only beyond $\approx 180\times$. At 2,560-d, projection plus a codec owns 8–63 $\times$ (DocVQA) and up to 105 $\times$ (InfoVQA).

These frontiers are in-sample. Held-out frontiers reproduced the axis composition within about 2 points, except at the extreme end.

VI. Why Compression Differs Across Datasets

The shipped rule chose much less compression on DocVQA than on InfoVQA. At $T = 0.95$ out of sample it chose $11.8\times$ against $72.7\times$ (ColPali) and $32\times$ against $355\times$ (ColEmbed). Two explanations predict this: DocVQA representations could be intrinsically more fragile under compression (compression sensitivity), or DocVQA’s retention could simply be harder to certify at a fixed query budget (certification uncertainty). The two are different quantities, and we measured them separately.

Intrinsic sensitivity is small. For the same pipeline, pool retention on DocVQA was within 0–2 points of InfoVQA. Only 0, 3 and 12 of 40 candidates (ColQwen2, ColQwen2.5, ColPali) differed beyond their intervals.

Certification uncertainty is large. The certification standard error at $n = 225$, $\text{sd}(c - Rb)/(\sqrt{225} \bar{b})$, was $2.1\text{--}2.6\times$ larger on DocVQA. It splits about evenly into two parts: a lower mean baseline (the denominator) and a larger per-query spread from near-boundary queries that both lose and gain under compression.

Relevance margins explain most of the gap (Fig. 2). Define a query’s relevance margin as the per-token score gap between its labelled page and the best other page. 56–58% of DocVQA queries fall in the two lowest margin quintiles, against 23–26% of InfoVQA queries. Two observational analyses close most of the certification-SE gap:

- reweighting InfoVQA to DocVQA’s margin distribution closes 61–81%;
- restricting DocVQA to its positive-margin half closes 84–120%. That half is a ceiling subset in which baseline nDCG@5 is ≈ 1 , so it overstates the match.

These analyses are observational, not causal. A residual remains, larger for ColPali. We therefore conclude that DocVQA receives less certified compression primarily because its quality is harder to certify, not because its representations are intrinsically less compressible, with residual uncertainty.

VII. Quality-Constrained Selection and Certification

The shipped rule is empirically safe but not certified. Across 20 random calibration/held-out splits per cell, the shipped 0.999-bound rule met its target on the held-out half in 355 of 360 choices for the 128-d encoders and SciFact, and in 119 of 120 for ColEmbed. The worst miss was 97.4% against 0.99. Choosing by the point estimate met the target in only 10–100% of splits per cell. Because the splits reuse one query pool per corpus, these counts are empirical summaries, not rates with population-level guarantees.

- Looser targets always buy at least as much compression (Fig. 3): in every dataset–encoder cell, the

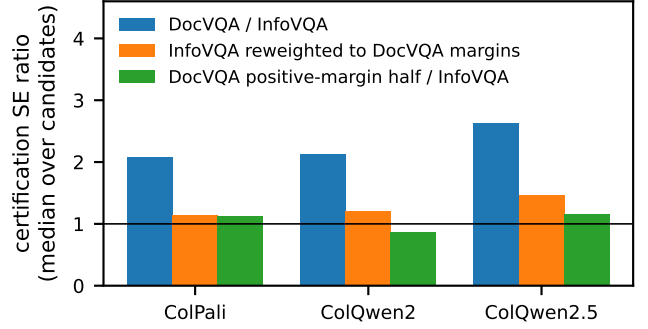


Fig. 2. Ratio of the $n = 225$ certification standard error between DocVQA and InfoVQA (median over 40 candidates; measured). Matching InfoVQA to DocVQA’s relevance-margin distribution, or restricting DocVQA to its positive-margin half, removes most of the gap. The analyses are observational, not causal.

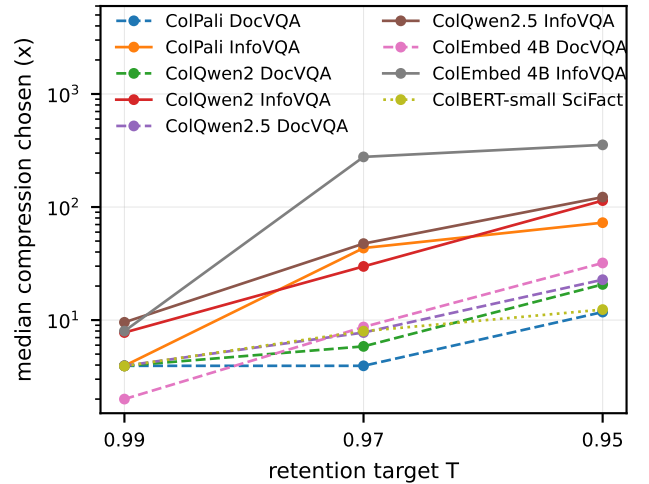


Fig. 3. Median compression chosen out of sample by the shipped rule at three retention targets (20 splits per cell; measured). Compression is dataset-dependent and never decreases as the target loosens. These are empirical results without a post-selection guarantee.

median chosen compression never decreased as the target loosened.

- The rule has no post-selection statement: in population resampling it missed the finite-pool target in up to 26% of draws at $n = 50$.

A certified alternative (Method C). We apply two established ideas. Exact p-values for the bounded mean in (1) come from testing by betting [20], with Ville’s inequality [21]. Each candidate family is walked in its pre-specified compression order by fixed-sequence testing [22], in the style of learn-then-test risk control [23], at level δ/F for F families. The smallest rejected configuration is deployed. Under i.i.d. queries, a fixed corpus and stages fitted without calibration queries, this gives

$$\Pr(\text{deploy} \wedge R_Q(\hat{k}) < T) \leq \delta \quad (2)$$

for the selected configuration $\hat{k}$.

Method C is a validated application of known methods, providing a conditional, finite-sample guarantee. It is not a new statistical method, it is not a guarantee for the shipped rule, and it does not hold for other corpora, growing corpora, or query distributions unlike the calibration pool.

Validation. In population resampling on ColPali and ColQwen2, and in a held-back confirmation frozen before loading ColQwen2.5 and SciFact, the bound held in every check:

- 27 of 27 held-back cells, with a maximum observed miss of 0.05% at $\delta = 0.05$;
- with ≤ 225 calibration queries, C deployed at $T = 0.95$ on all three held-back datasets (ColQwen2.5 DocVQA 7.8 $\times$, ColQwen2.5 InfoVQA 76.0 $\times$, SciFact 11.8 $\times$), and at $T = 0.97$ only on InfoVQA;
- it was never run on ColEmbed, for lack of per-query data.

The 0.99 limit. Consider any test valid at level δ , and contaminate a lossless candidate with an ε mass of total failures so that its retention equals T . Rejecting the null with probability at least $1/2$ then requires $n \geq \log(1/(2\delta))/(-\log(1-\varepsilon))$, where $\varepsilon = \beta(1-T)/(T+\beta(1-T))$ and β is the mean baseline score. At $T = 0.99$ and $\delta = 0.05$ this needs 250–392 queries on our pools, more after the δ/F correction. No distribution-free method can certify $T = 0.99$ with 225 or fewer calibration queries here, which is why a 0.95 result does not transfer to 0.99.

VIII. Search Cost, Memory and Scaling

Measured scaling. We tiled the ColQwen2 DocVQA corpus 1, 2, 4 and 8 times (500–4,000 identical duplicated pages, used for timing only) and timed exhaustive scans (Fig. 4). All six configurations were linear in stored vectors: the time at 4,000 pages was 0.98–1.08 of eight times the 500-page time, with $R^2 \geq 0.999$. The per-vector cost was similar across codecs (37–39 ms per million vectors per query) and only about 22% lower after halving the width. Vector count therefore remains the latency lever.

Two-tier search. Rescoring 50 candidates per query with the previous per-query loop cost ≈ 11.2 ms per query, flat in corpus size. The batched implementation cost 1.53–4.03 ms as the number of distinct candidate pages grew from 500 to 3,984. Its advantage therefore fell from 7.4 $\times$ to 2.8 $\times$ as overlap between queries fell; below that overlap it is untested. Extrapolating linearly (projected), the exhaustive hot stage exceeds the rescoring cost beyond $\approx 1,300$ pages. Two-tier indexing thus reduces RAM by about 30 $\times$ relative to int8 but does not change the linear scaling class.

Native compressed-code scoring. Scoring int8 codes directly with an available mixed-precision kernel, or binary codes with lookup tables, reproduced the current rankings up to float32 near-ties. It was 24–34 $\times$ and 28–30 $\times$ slower

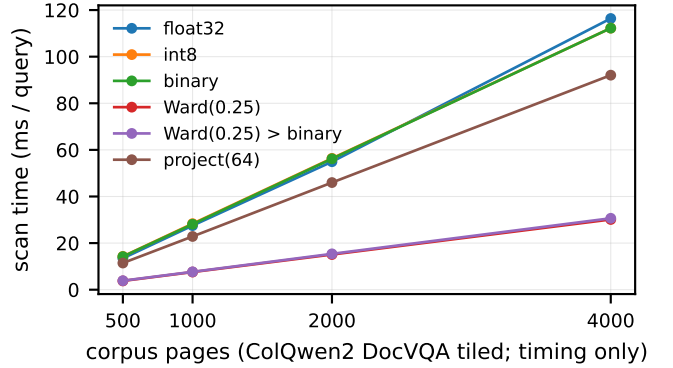


Fig. 4. Exhaustive scan time against corpus size on tiled ColQwen2 DocVQA (duplicated pages; timing only, no retrieval metric; measured on one laptop CPU). Time is linear in stored vectors over 500–4,000 pages.

than decode-then-BLAS. This is a negative result on this tested software stack (numpy and PyTorch on one CPU, with no compiled kernels), not a general property of compressed retrieval.

Projected resources (arithmetic on measured per-page bytes). Storage and RAM scale linearly in pages plus a small fixed shared state. At 1M pages, ColQwen2 projects to 383 GB in float32, 12 GB as binary codes and 3.1 GB merged to 1/4 then binarized. Under an 8 GB RAM budget, uncompressed 128-d indexes fit 16–23k pages, and merged binary fits 2.0–2.8M. These are projected resource figures, not demonstrated capability. Exhaustive search at that scale projects to several seconds per query on our laptop, and retrieval quality beyond 500 pages was not measured.

IX. Generalization Across Models and Corpora

Table II classifies the findings against the tested set. Our overall verdict is supported with scope limitations. Model dependence is concrete:

- PCA to $d/4$ loses 9–20 points at 128 dimensions but at most 0.9 points at $d/8$ for ColEmbed;
- plain int4 ranges from lossless (ColQwen) to collapse (ColEmbed);
- centred binary collapses on the text model, whose vectors occupy a narrow cone;
- a fixed merge radius merges about half as much on ColQwen as on ColPali.

Relative per-page budgets transferred where absolute thresholds did not. SciFact confounds the text model with a text corpus, so no claim about text corpora in general is supported.

X. Comparison With Prior Work

Related methods.

- Token reduction: Ward token pooling is prior work [8], and OptiVision’s merge stage implements it. Merging

TABLE II

Generalization within the tested set (four visual encoders across two widths, one text encoder, three corpora). “Replicated” means observed on every encoder and corpus where it was measured.

finding	status
vector count drives exhaustive search cost	replicated
relative merging beats pruning, preserves quality to $\approx 1/4$	replicated
int8 per vector near-lossless	replicated
centred int4 near-lossless (≤ 1.4 points)	replicated
factors do not multiply; PCA + coarse codecs antagonistic	replicated (ColEmbed: points only)
two-tier keeps quality, cuts RAM	replicated
certification difficulty drives dataset differences	replicated (DocVQA vs InfoVQA)
Method C’s conditional validity	replicated (4 encoders, 3 corpora; not ColEmbed)
dimension reduction	model-specific
plain int4; centred binary; 2-bit/binary strength	model-specific
absolute merge thresholds	do not transfer
amount of certifiable compression	dataset-specific
codec speed (decode-first)	implementation-specific

also outperforms pruning in Light-ColPali [9] and in a recent comparison of training-free methods on text [10]. Adaptive pruning and prune-then-merge methods target ColPali-family models [11]–[13].

- Codes and engines: residual compression and engines such as ColBERTv2/PLAID and EMVB [2], [15], [16] and fixed-dimensional encodings (MUVERA [17]) are established. So are binary multi-vector codes with float rescoring and shortlist-and-rescore designs [18], [19].
- Statistics: our selection procedure uses known statistical tools [20], [22], [23].

The one directly comparable setup. Our survey found a single published setup with the same checkpoints (ColPali v1.3 and ColQwen2 v1.0) and the same ViDoRe V1 subsets: Structural Anchor Pruning (SAP) [14].

- The disagreement: at about 1/10 of the vectors, our measured Ward merging retains 4–10 points more than SAP’s reported K-means baseline.
- Mechanism choices explain part of it: a controlled attribution on our data found that clustering and rescaling choices account for part of the gap.
- Evaluation is not the cause: aligning our evaluation with the official ViDoRe legacy and MTEB pipelines changed results by at most 0.65 points.
- What remains: on ColPali, random-pruning controls agree with SAP within 1.6 points. On ColQwen2 they disagree by up to 14 points before any compression method is applied, which points to an upstream difference (plausibly encoding settings).

The discrepancy is unresolved. We report it as an unresolved disagreement, not as evidence that either approach

is better or worse.

XI. Limitations and Open Questions

Scope.

- Four visual encoders across two widths (three of them 128-d), one text encoder, and three corpora. The visual corpora are 500-page ViDoRe V1 subsets.
- One query pool per corpus, so split-based rates are summaries, not population estimates.
- ColEmbed: aggregate results only, with GPU timing.
- Not covered: 4,096-dimensional or other model families; ViDoRe V2/V3; retrieval quality beyond 500 pages (an audit found lossy configurations losing 0–3 points from 100 to 500 pages); timing beyond 4,000 duplicated pages.

Implementation and statistics.

- Timing comes from one laptop with two CPU power states.
- Codec-speed and native-kernel results are specific to this implementation.
- The library does not yet persist centred int4 or 2-bit codecs, or the PCA basis.
- Method C assumes i.i.d. queries and a fixed corpus; the shipped rule’s success rates are empirical.
- Our literature survey covered multi-vector compression and retrieval engines, not the statistical-selection literature.

Open questions.

- the ColQwen2 discrepancy with SAP;
- the residual DocVQA certification gap;
- the cause of plain int4’s collapse at 2,560 dimensions;
- batched rescoring at very low candidate overlap;
- retrieval quality at large corpus sizes;
- whether compiled native kernels outperform decode-then-BLAS;
- sublinear candidate generation, which would be needed to change the scaling class.

XII. Discussion

Three lessons follow from the evidence, each within its tested scope:

- Compression cost and certification cost are distinct, and on DocVQA the second dominates. Reporting retention without the uncertainty of its estimate can make a dataset look less compressible than it is.
- Axes are not independent knobs. Byte factors do not multiply, and some pairings, notably projection followed by coarse codes, lose more than their parts.
- A smaller index is not a faster search in a decode-first implementation. Only fewer vectors reduced exhaustive cost here.

Each lesson is an empirical finding established on our tested set. Whether any is new relative to the complete literature is not established by this work.

XIII. Reproducibility

Every quantitative claim comes from frozen, pre-registered studies on the research-investigation branch of the OptiVision repository. The summary is in `reports/research/PAPER_SUMMARY.md`, with per-study reports:

- Q3: dataset differences;
- Q4: compression axes;
- Q5/Q6: native scoring;
- Q7, E7a, E7c: prior-work comparison;
- Q8: scaling;
- Q9: generalization;
- Q10: contribution classification;
- `reports/engineering/batched_rescore:` batched rescore.

The figures are regenerated from those reports by `paper/make_figs_research.py`. Encoded vectors are not stored in the repository; per-query result stores and scripts are.

XIV. Conclusion

OptiVision is a model-agnostic engineering framework for quality-constrained multi-vector compression. A systematic, pre-registered evaluation across four visual encoders, one text encoder and three corpora establishes several cross-model empirical findings:

- stored vector count governs exhaustive search cost;
- relative merging and per-vector int8 and centred int4 preserve quality;
- compression factors do not multiply, and the axes interact;
- dataset differences in certified compression stem mainly from certification difficulty.

It also identifies clear model-, dataset- and implementation-dependent boundaries: dimension reduction, low-bit codecs, the amount of certifiable compression, and codec speed. A known post-selection procedure gives a conditional finite-sample statement for the selected configuration, with a hard query-count limit at high targets. These findings are established within the tested scope; their novelty relative to the full literature is not.

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